

REDUCE-MAX EXCEEDS BACKLOG 2 ON TEN BAMBOOS

ABSTRACT. After normalizing total daily growth to 1, D’Emidio, Di Stefano, and Navarra conjectured that the greedy bamboo-trimming rule REDUCE-MAX, which cuts a tallest bamboo in each round, has backlog at most 2. Jasińska, Kuszmaul, and Lee disproved this conjecture with instances on 2000 and 2702 bamboos, and report that all of their counterexamples have between 1200 and 3000 bamboos and that similar steep distributions much smaller or larger yielded none. We give an exact ten-bamboo witness for which every tie-breaking rule has backlog

$$\frac{1878}{937} = 2 + \frac{4}{937} > 2.$$

The maximum is attained in round 60, and from the post-round-137 state onward the trajectory modulo permutation of six equal-rate bamboos is seven-periodic. The verification uses only integer arithmetic. As a consequence, positive-rate counterexamples exist for every number of bamboos $n \geq 10$.

1. THE COUNTEREXAMPLE

A bamboo instance has positive growth rates $r_1, \dots, r_n$ with $\sum_i r_i = 1$. Initially all heights are zero. In each round, every bamboo grows by its rate and one bamboo of maximum height is then cut to height zero. The backlog is the supremum of the maximum pre-cut height over all rounds. This greedy rule is called REDUCE-MAX.

D’Emidio, Di Stefano, and Navarra conjectured that REDUCE-MAX is 2-approximating with respect to total daily growth; under the normalization above, this is the backlog bound 2 [1, Conjecture 1]. Here 2-approximating is used in the sense of [1], where the property observed in their experiments is recorded as $M(E) < 2H$ with H the sum of the growth rates; it is that inequality which fails below. Nothing here bears on the ratio of $M(E)$ to the optimal backlog of the individual instance. Jasińska, Kuszmaul, and Lee disproved the conjecture, giving a 2000-bamboo example with backlog 2.0004 and a 2702-bamboo example yielding the stronger lower bound 2.076 [2, Section 4]. That disproof is theirs and no priority for it is claimed here: the contribution below is an exact witness with ten bamboos, and it does not improve the numerical lower bound 2.076. What it does settle is a question of size left open in [2], where it is reported that “all of our counterexamples have between 1200 and 3000 bamboos”, that “Examining similar

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steep distributions on much smaller or larger distributions did not yield examples with backlog greater than 2”, and that this is “likely because for smaller number of bamboos, the slower bamboos will not be numerous enough” [2, Section 4.1]. Theorem 1 shows that ten bamboos suffice, two orders of magnitude below that range, so no such lower limit on the number of bamboos holds.

Theorem 1. *For the rate vector*

$$r = \frac{1}{937}(313, 261, 131, 118, 19, 19, 19, 19, 19, 19),$$

every tie-breaking of REDUCE-MAX has backlog exactly 1878/937.

Proof. The numerator weights

$$w = (313, 261, 131, 118, 19, 19, 19, 19, 19, 19)$$

sum to 937. Scale all heights by 937. Let $H(t)$ be the integer post-cut state after round t , with $H(0) = 0$, and set

$$\hat{H}_i(t) = H_i(t-1) + w_i.$$

If bamboo j_t is cut in round t , then $H_{j_t}(t) = 0$ and $H_i(t) = \hat{H}_i(t)$ for $i \neq j_t$. Let $C(H)$ retain the first four coordinates and sort the last six in nondecreasing order.

A symbol $k \in \{1, 2, 3, 4\}$ means that bamboo k is cut. A symbol S means that a bamboo of maximum pre-cut height among coordinates 5, ..., 10 is cut. Since these six bamboos have the same rate, the resulting canonical state is independent of the choice among tied slow bamboos. For the indicated cut j_t , define the cross-rate gap

$$\delta_t = \hat{H}_{j_t}(t) - \max_{w_i \neq w_{j_t}} \hat{H}_i(t).$$

The following table is an exact certificate. The columns m , M , and Δ are, respectively, the minimum and maximum selected pre-cut heights and the minimum value of δ_t on the stated interval.

TABLE 1. Integer certificate for the greedy trajectory.

Rounds	Length	Cut word	m	M	Δ
1–6	6	121321	313	939	2
7–42	36	(421321) ⁶	708	939	4
43–60	18	SS214321SS21SS4321	817	1878	1
61–72	12	(214321) ²	522	1252	3
73–100	28	(S214321) ⁴	570	1252	23
101–108	8	SS214321	874	1252	27
109–137	29	(S214321) ⁴ S	684	1252	40
138–144	7	214321S	783	1252	40

For reference, the canonical post-cut states at the interval endpoints are

$$\begin{aligned}
C(H(0)) &= [0, 0, 0, 0 \mid 0, 0, 0, 0, 0], \\
C(H(6)) &= [0, 261, 262, 708 \mid 114, 114, 114, 114, 114], \\
C(H(42)) &= [0, 261, 262, 590 \mid 798, 798, 798, 798, 798], \\
C(H(60)) &= [0, 261, 262, 354 \mid 76, 95, 152, 171, 304, 323], \\
C(H(72)) &= [0, 261, 262, 354 \mid 304, 323, 380, 399, 532, 551], \\
C(H(100)) &= [0, 261, 262, 354 \mid 114, 247, 380, 513, 836, 855], \\
C(H(108)) &= [0, 261, 262, 354 \mid 114, 133, 266, 399, 532, 665], \\
C(H(137)) &= [313, 522, 393, 472 \mid 0, 133, 266, 399, 532, 665], \\
C(H(144)) &= [313, 522, 393, 472 \mid 0, 133, 266, 399, 532, 665].
\end{aligned}$$

All entries follow directly from the recurrence above. The positive values of Δ show that every indicated cut is globally greedy: for a digit, its rate class is a singleton; for S , the selected bamboo is maximal within the slow class by definition. The only ties for the global maximum occur in rounds

$$43, 44, 51, 52, 55,$$

and each is among equal-rate, equal-height slow bamboos. Hence every tie-breaking induces the same canonical trajectory.

The decisive comparisons in rounds 56–60 (the multipliers count rounds since the most recent cut) are

$$\begin{aligned}
56 : \quad & 56 \cdot 19 = 1064 > 9 \cdot 118 = 1062, \\
57 : \quad & 10 \cdot 118 = 1180 > 9 \cdot 131 = 1179, \\
58 : \quad & 10 \cdot 131 = 1310 > 5 \cdot 261 = 1305, \\
59 : \quad & 6 \cdot 261 = 1566 > 5 \cdot 313 = 1565, \\
60 : \quad & 6 \cdot 313 = 1878 = 2 \cdot 937 + 4.
\end{aligned}$$

Thus round 60 has pre-cut height $1878/937 > 2$. Table 1 shows that no other round through 144 has a larger height. Moreover, $C(H(137)) = C(H(144))$, and the intervening cut word is 214321 S . The canonical greedy evolution therefore repeats with period 7, with cycle peak 1252/937. Consequently the global backlog is exactly 1878/937, independently of tie-breaking. $\square$

Corollary 2. *For every integer $n \geq 10$, there is a positive-rate n -bamboo instance for which every tie-breaking of REDUCE-MAX has backlog greater than 2.*

Proof. Write $n = 10 + m$ with $m \geq 0$. The case $m = 0$ is Theorem 1. For $m \geq 1$, multiply the ten rates in the theorem by $999/1000$ and add m bamboos, each of rate $1/(1000m)$. The rates are positive and sum to 1.

Through round 60, every selected bamboo in the original execution has pre-cut height at least $313/937$, by Table 1. Inductively, before any new

bamboo has been cut, its height in a round $t \leq 60$ is at most

$$\frac{t}{1000m} \leq \frac{60}{1000} < \frac{999}{1000} \frac{313}{937}.$$

Thus no new bamboo can affect the canonical evolution of the original ten during these rounds. In round 60, the scaled first bamboo reaches

$$\frac{999}{1000} \frac{1878}{937} = 2 + \frac{2122}{937000} > 2.$$

The conclusion holds for every tie-breaking because the original ten-bamboo canonical trajectory is tie-independent. $\square$

Remark 3. The theorem gives a substantially smaller exact witness than the two examples displayed in [2], but it neither improves their lower bound 2.076 nor proves minimality. In particular, no claim is made about the existence of a counterexample with at most nine bamboos. The instance also lies outside the exhaustive search that supported [1, Conjecture 1]. Those experiments enumerated integer partitions of the total daily growth H , for $H \in \{5, 10, \dots, 35\}$ [1, Sections 4.1 and 4.2]. What excludes the rate vector of Theorem 1 is the bound on H : in integer form it has $H = 937$, so it was never enumerated there, even though its n is only 10.

REFERENCES

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